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MILITARY SPECIFICATION

TEST SET MARK 292 MOD 1

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification establishes the requirements for the procurement of Test Set Mk 292 Mod 1, LD 291414, which provides a means of checking the functions of Exploder Mk 19 Type. This test set is of one type and one class.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

Military

MIL-P-116	Preservation, Methods of
MIL-T-18404	Torpedoes, Environmental Requirements; General Requirements for
MIL-I-18997	Indicator, Pressure, Panel Mounted or Case Supported, General Specification

FSC 1355

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Naval Ordnance Systems Command

NAVORD LD 291414	Test Set Mark 292 Mod 1, and all Specifications and Drawings Listed Thereon
* NAVORD Dwg 1472706	Exploder Switch Base

PUBLICATIONS

Naval Ordnance Systems Command

OS 6341	Miscellaneous General Ordnance Requirements

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

National Bureau of Standards

Handbook H28	Screw Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Description. Test Set Mk 292 Mod 1 is a light-weight, portable unit capable of checking the performance of Exploder Mechanism Mk 19 Type. The test set is comprised of the following: aluminum case, hydraulic panel and applicable equipment, arming device simulator, two terminal ring assemblies, test lead assembly, electrical plate assembly, and an input power cable.

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This test set checks the operations of the exploder mechanism firing switch circuit, arming and disarming, ceiling switch, floor switch, switch base O-ring seal, and the exploder power supply output. The test set also provides unregulated test voltages of 150 volts direct current (dc) at 1.0 milliampere (ma); 24 volts dc at 2 amps; 6 volts dc at 2 amps; and 115 volts alternating current (ac), 400 Hertz (Hz) at 0.5 amp.

3.2 General requirements

3.2.1 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the test set shall be manufactured using the methods and procedures proposed for the production. The sample will be tested as specified in Section 4 herein and is for the purpose of determining, prior to production, that the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2.2 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the test set shall conform strictly to the detailed requirements indicated in the drawings and specifications listed in NAVORD LD 291414.

3.2.3 Finishes, coatings, construction, and dimensions. Finishes, coatings, construction, and dimensions of the test set shall be strictly in accordance with applicable drawings, specifications, and publications listed in NAVORD LD 291414.

3.2.4 Moisture, fungus, and corrosion. The test set shall be rendered moisture, fungus, and corrosion resistant in accordance with detailed requirements specified in the drawings and specifications listed in NAVORD LD 291414. When moisture, fungus, and corrosion resistance are not specified in these documents, the item shall be afforded such resistance in accordance with NAVORD OS 6341.

3.2.5 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in NAVORD LD 291414.

3.2.6 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

* 3.2.7 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

3.3 Definitions. Definitions applicable to the environmental requirements of this specification are defined in MIL-T-18404.

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3.4 Performance requirements and product characteristics. The test set shall meet the following performance requirements and product characteristics:

3.4.1 Power requirements. The test set shall operate from a 115 volts plus or minus 10 percent at 60 Hz plus or minus 5 percent input and shall generate working voltage as follows:

* 3.4.1.1 Voltage output. Voltmeters M1 and M2 shall be accurate within plus or minus 3 percent. The test set shall generate the following working voltages at the following working currents:

3.4.1.1.1 150 volts dc at 1.0 ma.

3.4.1.1.2 24 volts dc at 2.0 amps.

3.4.1.1.3 6 volts dc at 2.0 amps.

3.4.1.1.4 115 volts ac, 400 Hz at 0.5 amps.

* 3.4.2 Hydraulic system requirements. The hydraulic system, operated by a hand pump, shall be capable of supplying 610 to 620 psig water pressure for checking the operation of the exploder mechanism hydrostatic switches and O-ring seals. Gage accuracy shall be in accordance with MIL-I-18997.

3.4.3 Firing switch circuit. The test set shall show an indication of exploder firing switch closure in an armed exploder.

3.4.3.1 Calibration of firing switch circuit. The firing switch circuit shall operate with a resistance of 0.92 ohm.

3.4.3.2 Firing switch circuit resistor R16. The resistor plus associated wiring and connector test lead shall have a value within the range of 0.99 ohm minimum and 1.08 ohms maximum.

3.4.3.3 The resistance between terminals "A" and "E" of the terminal ring assembly shall not exceed 0.1 ohm maximum when TEST SELECTOR switch S3 is set at the FIRING SWITCH position.

3.4.4 Arming device simulator. The arming device simulator shall resist rotation in either direction with a frictional torque of 4 plus or minus 0.5 inch-pounds.

3.4.5 Floor and ceiling switches circuit. The floor and ceiling switches circuit shall operate as specified in 4.5.4.6.

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3.4.5.1 Power supply outputs. The voltages and polarities for the various mods of exploders shall be selected in accordance with 4.5.4.7.

* 3.4.6 Hydrostatic pressure. With 850 plus or minus 25 psig applied to the test set, there shall be no pressure drop after the system stabilizes in a maximum of 30 minutes at constant room temperature.

* 3.4.7 Gage accuracies. The 0-60 and 0-1000 psi gage accuracies shall be in accordance with MIL-I-18997.

3.5 Environmental requirements

* 3.5.1 Vibration. The test set shall be capable of withstanding vibration requirements as follows:

3.5.1.1 Vibration, transportation. When packaged and packed as specified in Section 5, the test set, after being subjected to the tests specified in 4.4.1.1, shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.5.1.2 Vibration, operating. The test set, less pack and packaging and self-mounted as specified in 4.4.1.2, shall be capable of withstanding vibration requirements as specified in 4.4.1.2 and, during and after, shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.5.2 Shock. The test set, mounted as specified in 4.4.2, shall be capable of withstanding shock requirements as follows:

3.5.2.1 Shock, transportation. When packaged and packed as specified in Section 5, the test set, after being subjected to the tests specified in 4.4.2.1, shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.5.2.2 Shock, nonoperating or ready. The test set, less pack and packaging and self-mounted as specified in 4.4.2.2, shall be capable of withstanding shock requirements as specified in 4.4.2.2 and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.5.3 Temperature. The test set shall be capable of withstanding temperature conditions specified in 4.4.3 and shall meet the performance requirements and product characteristics of this specification.

3.5.4 Humidity. The test set shall be capable of withstanding humidity conditions specified in 4.4.4 and thereafter shall show no corrosion or

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damage and shall meet the performance requirements and product characteristics of this specification.

3.5.5 Tropical environment. The test set shall be capable of withstanding the tropical conditions specified in 4.4.5 and thereafter shall show no fungus growth, corrosion, or other damage and shall meet the performance requirements and product characteristics of this specification.

3.6 Cleanup. Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges, as well as rosin flash, shall be removed.

3.7 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this publication. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, markings of parts and assemblies, plating, painting, riveting, welding, brazing, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the test set shall be classified as follows:

(a) Preproduction inspection (paragraph 4.4)

(b) Periodic production inspection (paragraph 4.4)

* (c) Quality conformance inspection (paragraph 4.5)

4.3 Sampling.

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4.3.1 Preproduction sample. Unless otherwise specified by the procuring activity, a preproduction sample of five test sets manufactured in accordance with 3.2.1 shall be subjected to the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the test sets by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Periodic production sampling. Periodic production samples shall be randomly selected from units that have satisfactorily passed the quality conformance inspections of 4.5. These samples shall be subjected to the tests of 4.4 at an activity designated by the procuring activity. Unless otherwise specified by the procuring activity, periodic production sampling shall be as follows:

Step 1. Upon initiation of production randomly select one unit from each group of 10 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 2.

Step 2. Randomly select one unit from each group of 25 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 3.

Step 3. Randomly select one unit from each group of 50 units. Continue Step 3 for the balance of the contract or until a defective unit is found.

NOTE: Failure in Steps 2 or 3 requires return to Step 1.

4.3.3 Acceptance sampling. The tests and examinations detailed in 4.5 shall be conducted on each test set to be delivered; therefore, there shall be no sampling for acceptance. Defective units shall be rejected.

* 4.4 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.5, shall be subjected to the following tests and examinations. When specified in the contract or order, certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

4.4.1 Vibration. The test set shall be mounted and vibration tested as specified in 4.4.1.1 and 4.4.1.2. At the conclusion of the tests, the test set shall meet the performance and other requirements of this specification.

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4.4.1.1 Vibration, transportation. The test set, packaged and packed as specified in Section 5, shall be vibration tested as follows:

4.4.1.1.1 The unit shall be vibrated at 1 g input in such a manner as to discover the frequency (or frequencies) producing maximum transmissibility within the frequency range of 5-60 Hz.

4.4.1.1.2 The unit shall be fastened in suitable fashion (maximum permissible motion freedom approximately 1/32 inch) to the table of a vibration testing device producing an essentially sine wave motion. After determining frequency (or frequencies) where maximum transmissibility occurs, vibrate the test set at this frequency (or frequencies) for a period of one hour. This test is to be applied to each of the three principal orthogonal axes.

4.4.1.2 Vibration, operating. The test set shall be mounted to a vibration device in a manner simulating its mounting in actual use. Energize the test set. The tests of 4.4.1.2.1, 4.4.1.2.2, and 4.4.1.2.3 shall be accomplished in each of the three mutually perpendicular axes. The unit shall operate satisfactorily without repair while performing its principle functions and shall meet the performance requirements of this specification.

* 4.4.1.2.1 Exploratory vibration test. The unit shall be vibrated at frequencies from 10 to 33 Hz with a total excursion of 0.010 inch plus or minus 0.002 inch. The change in frequency shall be in one Hz increments every 15 seconds. All resonant frequencies shall be noted.

4.4.1.2.2 Variable frequency test. The test set shall be vibrated for a minimum of 5 minutes in each integral frequency range below.

Each integral frequency range shall be varied uniformly from one limit to the other and back to the lower limit in a maximum time of 1/2 minute.

Frequency (Hz)	Total Excursion (inch)
5 to 15	0.030 ± 0.006
* 16 to 25	0.020 ± 0.004
26 to 33	0.010 ± 0.002

* 4.4.1.2.3 Endurance test. The test set shall be vibrated for a minimum of 2 hours at the resonant frequencies noted in 4.4.1.2.1. If no resonant frequencies have been noted, this test shall be accomplished at 33 Hz with a total excursion of 0.010 ± 0.002 inch.

4.4.2 Shock. The test set, mounted as in 4.4.2.1 and 4.4.2.2, shall be subjected to the shock testing specified in those paragraphs. At the

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conclusion of this test, the test set shall meet the performance and other requirements of this specification.

4.4.2.1 Shock, transportation. The test set, packaged and packed as specified in Section 5, shall be subjected to the rough handling tests specified in Section 4 of MIL-P-116 for a container not exceeding 200 pounds.

4.4.2.2 Shock, nonoperating or ready condition. The test set, less pack and packaging, shall be mounted securely to a shock machine and subjected to shock tests of 15 g's on all axes for not less than 8 milliseconds pulse duration, and approximating a 1/2 sine wave. At the conclusion of this test, the test set shall meet the performance and other requirements of this specification.

4.4.3 Temperature

4.4.3.1 Low temperature test. With the unit nonoperating, lower the temperature to minus 65 degrees F (minus 54 degrees C) and allow the unit to stabilize at this temperature. Maintain the temperature for 48 hours plus or minus 5 hours. At the end of this period raise the temperature to 77 degrees F (25 degrees C) plus or minus 5 degrees, allow the unit to stabilize, and perform the tests of 4.5. The unit shall pass these tests. Lower the temperature to 40 degrees F (5 degrees C) plus or minus 5 degrees and allow the unit to stabilize. Maintain the temperature for 30 minutes and, during this period, perform the tests of 4.5.

4.4.3.2 High temperature test. With the unit nonoperating, raise the temperature to 160 degrees F (71 degrees C) with 90 percent relative humidity. Allow the unit to stabilize at this temperature and maintain the temperature for 48 hours plus or minus 5 hours. At the end of this period lower the temperature to 77 degrees F (25 degrees C) plus or minus 5 degrees with 90 percent relative humidity; allow unit to stabilize and perform tests of 4.5. The unit shall pass these tests. Raise the temperature to 130 degrees F (55 degrees C) plus or minus 5 degrees with 90 percent relative humidity; allow the unit to stabilize and maintain for 30 minutes. The unit shall pass the tests of 4.5 while still at this temperature.

4.4.4 Humidity test. While it is in a nonoperating condition, dry the unit for 24 hours at 130 degrees F (55 degrees C). Place the unit in a humidity chamber and maintain the relative humidity between 90 and 95 percent at temperatures above 35 degrees F during the following temperature cycle:

Lower the temperature to minus 40 degrees F and hold for 2 hours. At the end of this period raise the temperature to 160 degrees F (71 degrees C).

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This temperature shall be reached in the chamber within 2 hours. Hold at this temperature for a period of two hours. Lower the temperature to 90 degrees F (32 degrees C) at the rate of not less than 10 degrees F per hour. While at 90 degrees F the unit shall meet the requirements of 4.5 and shall be operated only long enough to perform the tests. Lower the temperature to 70 degrees F (21 degrees C). This temperature shall be reached in the chamber within 2 hours. While at this temperature the unit shall meet the requirements of 4.5. The temperature cycle of this humidity test shall be satisfactorily completed four times.

4.4.5 Tropical environment. The panels shall be removed from their cabinets so that all parts and circuit elements are exposed. All components of the unit shall be placed in an incubator where conditions are maintained at 86 degrees F (30 degrees C) plus or minus 9 degrees F and 90 to 100 percent relative humidity with condensation at atmospheric pressure. The air should be relatively stagnant except for slight ventilation by convection caused by leaks through the chamber. Parts shall be inoculated with a mixture of fungi spores of the following species:

Penicillium Luteum
Aspergillus Niger
Aspergillus Flavus
Trichoderma
Lignorum T-1 USDA

Spores shall be mixed with distilled water and approximately 1/2 ounce of the mixture shall be atomized into the incubator above the parts but not directly on them. A fresh mixture must be used for each test inoculation.

The inoculated unit shall be incubated for a period of 28 days. At the end of this period, the parts shall be visibly examined for evidence of fungus growth. Any visible growth will be cause for rejection. The unit shall be reassembled and must meet the performance requirements of this specification. Drying of electrical connectors to prevent short circuits due to moisture formation is permissible prior to performance testing. Drying may be accomplished with pressurized air or by wiping. The unit shall not be dried by baking.

4.4.6 Functional tests. At the conclusion of the tests specified in 4.4.1.1 and 4.4.2.1 above, functional and other tests, as applicable, shall be conducted on the test set to determine freedom from operational malfunction and compliance with all requirements of Section 3.

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* 4.5 Quality conformance inspection. Each test set delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.5.1 Test conditions. Unless otherwise specified, the test set shall be subjected to quality conformance tests under the following conditions:

4.5.1.1 Temperature: Room ambient 65 degrees F to 95 degrees F (18 degrees C to 35 degrees C).

4.5.1.2 Altitude: Normal ground.

4.5.1.3 Vibration: None.

4.5.1.4 Humidity: Room ambient to 95 percent relative maximum.

4.5.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity and, when definite requirements are not specified, shall be of sufficient accuracy and quality to permit performance of the required quality conformance tests.

4.5.3 Test equipment

4.5.3.1 An ac voltmeter, 0 to 150 volts, 24 to 400 Hz, having an accuracy of plus or minus 0.5 percent or better.

4.5.3.2 A dc voltmeter, 0 to 150 volts, having an accuracy of plus or minus 0.5 percent or better.

4.5.3.3 A dc ammeter, 0 to 2 amperes, having an accuracy of plus or minus 3 percent or better.

4.5.3.4 A dc milliammeter, 0 to 1 milliampere having an accuracy of plus or minus 3 percent or better.

4.5.3.5 An ac ammeter, 0 to 1/2 ampere 400 Hz, having an accuracy of plus or minus 3 percent or better.

4.5.3.6 Resistance bridge or other resistance measuring device capable of measuring one ohm with an accuracy of plus or minus 1 percent or better.

4.5.3.7 Hydraulic gage calibration equipment, 0 to 1000 psi.

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4.5.4 Test procedure

NOTE: During the following tests, all switches should be in the OFF position unless otherwise indicated. (MOD SELECTOR switch S2 has no OFF position. On the 0, 1, 3, or 14 position, the dynamotor runs when the power is on.)

4.5.4.1 AC voltmeter M1 accuracy test. The accuracy of 400 Hz voltmeter M1 shall be tested at the 115 volt point, utilizing the output of motor-generator MG1. Connect an external 400 Hz ac voltmeter having an accuracy of better than 0.5 percent to output binding posts E7 and E8. With 115 volt, 60 Hz ac power supplied to test set, set POWER ON-OFF switch S1 to the ON position. To operate the motor-generator, set MOD SELECTOR switch S2 to either the 0, 1, 3, or 14 position and adjust resistor R5 until a reading of 115 volts is obtained on the external voltmeter. Voltmeter M1 shall read within plus or minus 3 percent of the external meter. Set POWER ON-OFF switch S1 to the OFF position.

4.5.4.2 DC voltmeter M2 accuracy test. The accuracy of dc voltmeter M2 shall be tested at the 150 volt point by removing 3/16 ampere fuse F2 from fuseholder XF2, placing DC METERING switch S9 in its 150 volt position, and applying an external source of 150 volts dc to binding posts E3 and E4, as measured by an external dc voltmeter having an accuracy of 0.5 percent or better. The reading on dc voltmeter M2 shall be within plus or minus 3 percent of the external voltmeter.

4.5.4.2.1 The accuracy of dc voltmeter M2 shall be tested at the 24 and 6 volt points by removing 2 ampere fuse F3 from fuseholder XF3, placing DC METERING switch S9 in the 24 or 6 volt position, as applicable, and applying an external source of 24 or 6 volts dc, as applicable, to binding posts E5 and E6, as measured by an external dc voltmeter having an accuracy of 0.5 percent or better. The reading on dc voltmeter M2 shall be within plus or minus 3 percent of the external voltmeter.

4.5.4.3 Firing switch indicator lamp circuit. Place SELECTOR switch S3 in the FIRING SWITCH position and DC METERING switch S9 in the 100 volt position. Set POWER ON-OFF switch S1 to the ON position. Depress and then release CALIBRATE pushbutton switch S5 and note if FIRING SWITCH indicator lamp DS 3 is on or off; in either case, adjust as follows:

4.5.4.3.1 If FIRING SWITCH indicator lamp DS3 does not light, rotate PULSE ADJUST potentiometer R11 slightly in a clockwise direction. Depress and then release CALIBRATE pushbutton switch S5 and note if indicator lamp is on or off. If lamp still does not light, continue to rotate PULSE ADJUST

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potentiometer R11 in a clockwise direction and depress and release CALIBRATE pushbutton switch S5, allowing a 5 to 10 second interval between repetitions until FIRING SWITCH indicator lamp DS3 just lights.

NOTE: The lamp must light with the most minute adjustment of PULSE ADJUST potentiometer R11. After the adjustment is made, to ensure that FIRING SWITCH indicator lamp DS3 still goes on, depress and release RESET S4 and CALIBRATE S5 pushbutton switches again.

4.5.4.3.2 If FIRING SWITCH indicator lamp DS3 does light, rotate PULSE ADJUST potentiometer R11 slightly in a counterclockwise direction, then depress and release RESET pushbutton S4. Depress and then release CALIBRATE pushbutton switch S5 and note if the indicator lamp lights. If the lamp is still lit, continue to rotate potentiometer in a counterclockwise direction. Depress and release RESET pushbutton and CALIBRATE pushbutton switches, in that order, until the lamp just goes off, and then rotate the potentiometer until the lamp just lights. See above note.

4.5.4.4 Firing switch circuit resistance and resistor R16. Set POWER ON-OFF switch S1 to the OFF position. Connect a resistance measuring device, specified in 4.5.3.6, across the special two-prong connector and depress CALIBRATE pushbutton switch S5. Resistance of the firing switch circuit and R16 shall be within the range of 0.99 ohm MINIMUM to 1.08 ohms MAXIMUM.

4.5.4.5 Arming device simulator test. To ensure that the exploder arming motor and associated gearing function properly, the arming device simulator shall be tested in accordance with 3.4.4.

4.5.4.5.1 Centering the coupler. For this test the aligning pin farthest from the bore rod hole will be referred to as the "X" pin. Prior to conducting this test the coupler notch shall be pointed toward the "X" pin and the centerline of the notch shall be in alignment with the center of the "X" pin. As viewed from the underside of the simulator, a counterclockwise rotation of the coupler through an angle of 47 degrees, 43 minutes, plus or minus 0 degrees, 15 minutes, shall cause the leading edge of the yellow band to be in alignment with the index line of the viewing window. (The leading edge of the yellow band is defined as the edge which is farthest from the green band.) An additional counterclockwise rotation of the coupler 84 degrees, 15 minutes, plus 0 degrees, minus 0 degrees, 15 minutes, shall cause the leading edge of the red band to be in alignment with the index line on the viewing window. (The leading edge is the edge which first appears in the viewing window.)

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4.5.4.6 Floor and ceiling switches circuit test

NOTE: This test and all subsequent tests shall be accomplished with both terminal ring assemblies. With EXPLODER TEST SELECTOR switch S3 set at HYDROSTAT and POWER ON-OFF switch S1 set at the ON position, a short applied to the terminal ring assembly between pins "A" and "E" shall cause yellow indicating lamp DS2 to light.

* 4.5.4.7 Power supply outputs. With the test set connected to a steady source of 115 volts plus or minus 10 percent, 60 Hertz plus or minus 5 percent, and POWER ON-OFF switch S1 set at the ON position, voltage readings at rated current shall be taken from the binding posts of the test set and shall be as follows:

E3 to E4	150 volts dc plus or minus 20 percent at 1 ma.
E5 to E6 (S2 at 8 position)	6 volts dc plus 20 percent minus 60 percent at 2 amps.
E5 to E6 (S2 at all other positions)	24 volts dc plus 20 percent minus 30 percent at 2 amps.
E7 and E8	115 volts ac plus 20 percent minus 30 percent, 400 Hz plus or minus 20 percent at 0.5 amp.

Voltage between binding posts E5 and E6 shall be 24 volts dc for all positions of MOD SELECTOR switch S2, except 8, which is 6 volts dc. For Exploder Mk 19 Mods 5, 9, 12, and 13 only, when ARMING switch S6 is in the DISARM position, voltage at the terminal ring assembly between pins "A" and "C" shall be 24 volts dc with pin "A" positive and pin "C" negative. With arming switch S6 in the ARM position, pin "A" then becomes negative and pin "F" shall read 24 volts dc positive.

4.5.4.7.1 With MOD SELECTOR switch S2 set at Mods 4, 6, 7, 10, or 11, the voltage measured between pins "A" and "C" shall read 24 volts dc. With arming switch S6 set to DISARM position, pin "A" shall be negative and pin "C" positive. With ARMING switch S6 in the "ARM" position, pin "A" then becomes positive and pin "F" negative 24 volts dc.

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4.5.4.7.2 With MOD SELECTOR switch S2 set at 8, the voltage measured between pins "A" and "C" shall be 6 volts dc. With ARMING switch S6 set to DISARM position, pin "A" shall be positive and pin "C" negative. With ARMING switch S6 in ARM position, pin "A" then becomes negative and pin "F" positive 6 volts dc.

4.5.4.7.3 With MOD SELECTOR switch S2 set to Mods 0, 1, 3, or 14, panel voltmeter M1 shall indicate an output of 115 volts ac (pin) "B" shall be negative) at 400 Hz. With ARMING switch S6 in ARM position and MOD SELECTOR switch S2 in Mod 1, 3, or 14 position, the external voltmeter shall indicate an output of 115 volts ac at 400 Hz between pins "A" and "F". With ARMING switch S6 set at DISARM position the output of 115 volts ac at 400 Hz shall appear between pins "A" and "C".

4.5.4.7.4 With TEST SELECTOR switch S3 set at POWER SUPPLY position, the output between pins "A" and "B" shall be 150 volts dc. With MOD SELECTOR switch S2 set at Mods 4, 5, 6, 7, 9, 10, 11, 12, and 13 positions, pin "A" shall be negative. With MOD SELECTOR switch S2 set at Mod 0, 1, 3, or 14, and ARMING switch S6 set at OFF, the output measured between pins "A" and "B" by means of an external ac voltmeter shall be 108 volts ac at 400 Hz.

4.5.4.7.5 With DC METERING switch S9 turned successively to indicated settings the outputs indicated below shall be observed at 0 to 40/200 volt dc meter M2 and verified at test points to within plus or minus 10 percent by an external meter having a two percent accuracy:

Position of Switch S9	Readings on Meter M2	Test Points
Exploder power supply	150 volts	Pins "A" and "E" apply external power source 150 volt dc.
150 volt dc	150 volts	Output between bind-posts E3 and E4 (E3 positive)
115 volt dc	115 volts	No external test points
100 volt dc	100 volts (S3 at FIRING switch position)	Output between positive side of two prong connector and binding post E4
24 or 6 volt dc	6 volt dc with S2 set at 8; 24 volt dc with S2 set at all other positions	Output between binding posts E5 and E6 (E6 negative)

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* 4.5.4.8 Hydraulic pressure test. Mount a dummy switch base, having the dimensions of the Exploder Switch Base 1472706 (which delineates switch base assembly typical for Exploder Mk 19 Type), and an O-ring seal in the cavity of the test set. Supply a hydraulic pressure of 850 plus or minus 25 psig. There shall be no pressure drop after the system has stabilized in a maximum time of 30 minutes at constant room temperature.

4.5.4.9 Firing switch circuit test. With POWER ON-OFF switch S1 set to the ON position and test selector switch S3 set at firing switch position, firing switch indicator lamp DS3 shall light when a 0.92 ohm precision resistor is instantaneously connected across the terminals of the special two-prong connector attached to the test set.

4.5.4.9.1 Set POWER ON-OFF switch S1 to the OFF position. Connect resistance measuring device specified in 4.5.3.6 to pins "A" and "E" of the terminal ring assembly. The resistance measured shall be 0.1 ohm or less.

4.5.4.10 Visual and dimensional examination. Each item shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.5.4.11 Packaging, packing, and marking. Examine the packaging, packing, and marking to determine conformance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

* 5.1.1 Level A

5.1.1.1 Cleaning, drying, and preservative application. Not applicable.

* 5.1.1.2 Unit packaging. Unit packaging of the test set shall be in accordance with Method IIe of MIL-P-116. Each test set shall be packaged as a unit.

5.1.1.3 Intermediate package. Not applicable.

5.1.2 Level B. Same as 5.1.1.2.

5.1.3 Level C

5.1.3.1 Cleaning, drying, and preservative application. Not applicable.

5.1.3.2 Unit packaging. Unit packaging of the test set shall be in accordance with Method III of MIL-P-116. Each test set shall be packaged as a unit.

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5.1.3.3 Intermediate package. Not applicable.

5.2 Packing

5.2.1 Level A

* 5.2.1.1 Exterior containers. The test set, packaged as described in 5.1.1.2, shall be packed in a wood box in accordance with PPP-B-621, Class 2 or PPP-B-601, overseas type or in a cleated fiberboard box in accordance with PPP-B-591, overseas type.

* 5.2.2 Level B. Not applicable.

5.2.3 Level C. The test set, packaged in accordance with 5.1.3.2, shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Levels A, B, or C cushioning. The unit package or pack shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices to ensure that the test set will meet all the requirements of Section 3 after being tested in accordance with 4.4.1.1 and 4.4.2.1. At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent.

* 5.4 Marking. In addition to the markings required by contract or order, unit packages, intermediate packages (when used), and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended use. Test Set Mk 292 Mod 1 is intended for use in testing Exploder Mechanism Mk 19 Type.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Any exception to this specification.

(c) Required level of preservation and packaging.

(d) Required level of packing.

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(e) Assigned activity for preproduction and periodic production inspections.

(f) That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

6.2.1 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in MIL-P-116 and to 4.4.7 of this specification.

* 6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DIDs
4.4	Certified inspection data	-

DIDs (Data Item Descriptions/DD Forms 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDs when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

Custodian:
Navy - OS

Preparing Activity
Navy - OS
Project No. 1355-N329

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-T-22393B(OS) Test Set Mark 292 Mod 1

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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